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INTRODUCTION

Megafoods of Northern Asia

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ABSTRACT

'Megafood' refers to very-large-scale, high-energy catastrophic outbursts of water from natural and artificial reservoirs. For example, such outbursts may be caused by the collapse of a natural dam, or by the rapid introduction into a large waterbody of significant amounts of rocks and/or ice. Megafoods have periodically occurred in Northern Asia during the Pleistocene and probably during the Holocene as well. This special issue of *International Geology Review* presents a collection of papers dealing with various aspects of this fascinating topic.

KEYWORDS

northern Asia; lake; river; megafood; tsunami; jokulhlaup

'Megafood' refers to very-large-scale, high-energy catastrophic outbursts of water from natural and artificial reservoirs. For example, such outbursts may be caused by the collapse of a natural dam, or by the rapid introduction into a large waterbody of significant amounts of rocks and/or ice. Especially large megafoods were associated with the collapse of the Pleistocene ice dam holding back glacial Lake Missoula in the north-western USA. Another example of megafooding is a jokulhlaup, named for Icelandic large-scale glacier-associated flooding due to a volcanic eruption or other volcanothermal activity beneath an ice sheet. Megafoods involve enormous volumes of water (typically hundreds to thousands of cubic kilometres) that are discharged at extremely high rates. The flows commonly exceed one million cubic metres per second. Effects of the strongest megafoods are difficult to imagine because modern civilization has never been faced with such catastrophes, though ancient people could have been the victims of some of them. However, two landslide-generated megatsunamis have been evidenced in the twentieth century – the Lituya Bay, Alaska (9 July 1958), and Vajont artificial reservoir, Italy (9 October 1963).

Megafoods have periodically occurred in Northern Asia during the Pleistocene and probably during the Holocene as well. This special issue of *International Geology Review* presents a collection of papers dealing with various aspects of this fascinating topic.

The special issue opens with the review paper by Komatsu *et al.* (2016) that considers, in varying degrees of detail, 16 megafooding examples and associated palaeolakes from all over Northern Asia (see localities

of all megafoods considered in the special issue in Figure 1 of Komatsu *et al.* 2016). The great majority of these megafooding examples are associated with glacially dammed lakes, but one event is related to subglacial volcanic eruptions. This review is followed by another review paper and also a research paper, both by Batbaatar and Gillespie (2016a, 2016b), who provide very detailed considerations of megafooding events along the Maly Yenisei River. The potential source of the flooding was the glacially dammed palaeo-Darhad lake of northern Mongolia. The next two papers deal with different aspects of megafooding events in the Russian Altai mountain region. Baryshnikov *et al.* (2016) provide new information on the geology, geomorphology and geochronology of flooding from Lake Teletskoye via the Biya River. Bohorquez *et al.* (2016) use geological and geomorphological data as boundary conditions for computer modelling of the drainage of the Kuray–Chuja glacially dammed palaeolake following an instantaneous dam collapse. The special issue is completed by paper that presents the hypothesis of a landslide-generated megatsunami at Lake Baikal and its associated megafooding to the Lena River via the Goloustnaya, Buguldeika and Manzurka palaeoriver system (Ivanov *et al.* 2016a). This is followed by a critique of the hypothesis by Mats (2016) and a reply to that critique by Ivanov *et al.* (2016b).

We hope that this special issue will stimulate future studies of paleoflooding events in Northern Asia and worldwide. It may even raise awareness about potential catastrophes in association with the disruption of artificial dams and landslide-generated tsunamis.

Disclosure statement

No potential conflict of interest was reported by the authors.

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